

Администрирование сетевых подсистем

Лабораторная работа №10 — Расширенные настройки SMTP-сервера

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Цели и задачи работы

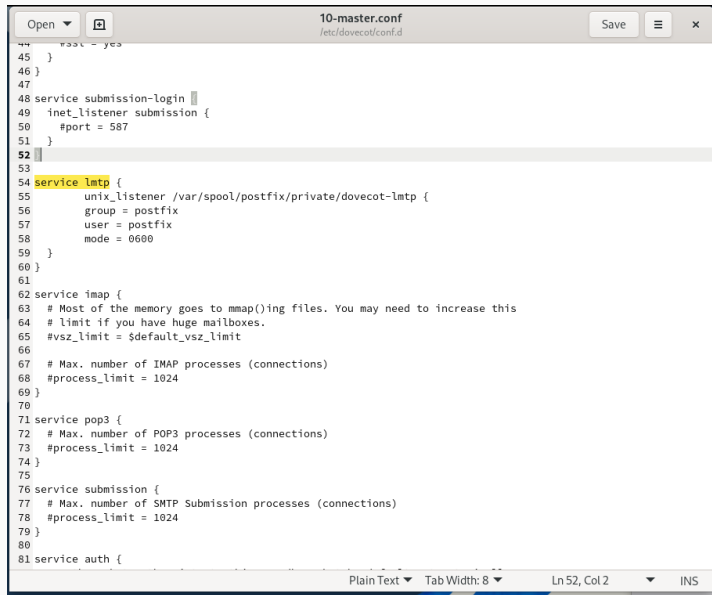
Цель лабораторной работы

Приобрести практические навыки конфигурирования SMTP-сервера в части настройки аутентификации и защищённой передачи данных.

- Настроить доставку почты через LMTP (Postfix → Dovecot)
- Включить SMTP-аутентификацию (SASL через Dovecot)
- Настроить SMTP over TLS (STARTTLS) и порт submission 587
- Проверить работу через консольные тесты и почтовый клиент
- Зафиксировать изменения в provisioning-окружении Vagrant

Выполнение лабораторной работы

Добавление LMTP в список протоколов

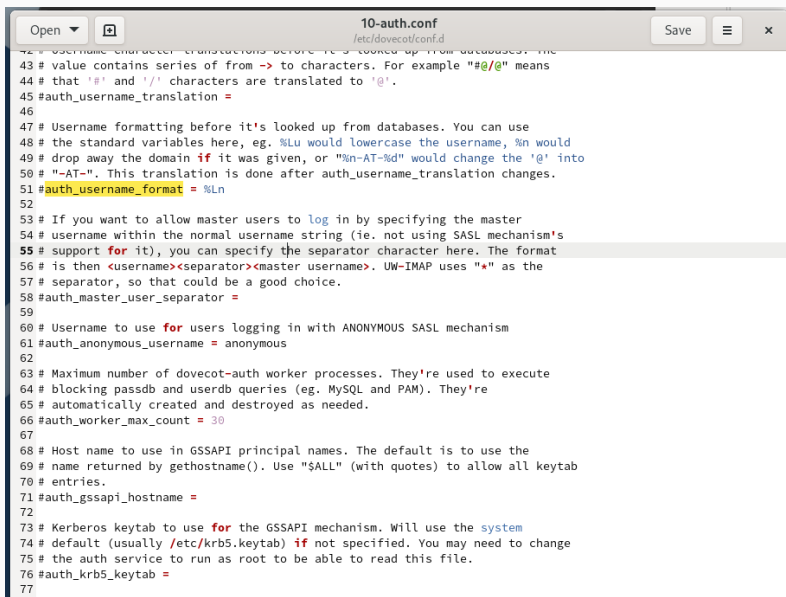


```
44 }
45 }
46 }
47
48 service submission-login {
49   inet_listener submission {
50     #port = 587
51   }
52 }
53
54 service lmt {
55   unix_listener /var/spool/postfix/private/dovecot-lmt {
56     group = postfix
57     user = postfix
58     mode = 0600
59   }
60 }
61
62 service imap {
63   # Most of the memory goes to mmap()ing files. You may need to increase this
64   # limit if you have huge mailboxes.
65   #vsz_limit = $default_vsz_limit
66
67   # Max. number of IMAP processes (connections)
68   #process_limit = 1024
69 }
70
71 service pop3 {
72   # Max. number of POP3 processes (connections)
73   #process_limit = 1024
74 }
75
76 service submission {
77   # Max. number of SMTP Submission processes (connections)
78   #process_limit = 1024
79 }
80
81 service auth {
```

Open ▾  10-master.conf
etc/dovecot/conf.d Save ≡ ×

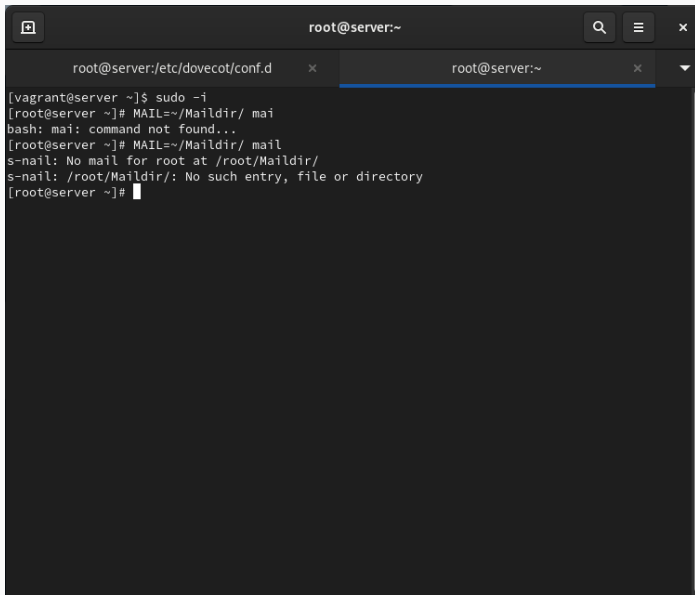
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Настройка сервиса LMTP и unix-сокета



```
42 # Username character translations before it's looked up from databases. The
43 # value contains series of from -> to characters. For example "#@/@" means
44 # that '#' and '/' characters are translated to '@'.
45 #auth_username_translation =
46
47 # Username formatting before it's looked up from databases. You can use
48 # the standard variables here, eg. %Lu would lowercase the username, %n would
49 # drop away the domain if it was given, or "%n-AT-%d" would change the '@' into
50 # "-AT-". This translation is done after auth_username_translation changes.
51 #auth_username_format = %Ln
52
53 # If you want to allow master users to log in by specifying the master
54 # username within the normal username string (ie. not using SASL mechanism's
55 # support for it), you can specify the separator character here. The format
56 # is then <username><separator><master username>. UW-IMAP uses "*" as the
57 # separator, so that could be a good choice.
58 #auth_master_user_separator =
59
60 # Username to use for users logging in with ANONYMOUS SASL mechanism
61 #auth_anonymous_username = anonymous
62
63 # Maximum number of dovecot-auth worker processes. They're used to execute
64 # blocking passdb and userdb queries (eg. MySQL and PAM). They're
65 # automatically created and destroyed as needed.
66 #auth_worker_max_count = 30
67
68 # Host name to use in GSSAPI principal names. The default is to use the
69 # name returned by gethostname(). Use "$ALL" (with quotes) to allow all keytab
70 # entries.
71 #auth_gssapi_hostname =
72
73 # Kerberos keytab to use for the GSSAPI mechanism. Will use the system
74 # default (usually /etc/krb5.keytab) if not specified. You may need to change
75 # the auth service to run as root to be able to read this file.
76 #auth_krb5_keytab =
77
```

Формат имени пользователя для аутентификации



```
root@server:~  
root@server:/etc/dovecot/conf.d x root@server:~ x  
[vagrant@server ~]$ sudo -i  
[root@server ~]# MAIL=~/.Maildir/ mai  
bash: mai: command not found...  
[root@server ~]# MAIL=~/.Maildir/ mail  
s-nail: No mail for root at /root/.Maildir/  
s-nail: /root/.Maildir/: No such entry, file or directory  
[root@server ~]#
```


Доставка сообщения: анализ логов

```
** (gedit:3411): WARNING **: 11:19:25.774: Set document metadata failed: Setting attribute m
etadata::gedit-spell-language not supported

** (gedit:3411): WARNING **: 11:19:25.779: Set document metadata failed: Setting attribute m
etadata::gedit-encoding not supported

** (gedit:3411): WARNING **: 11:19:38.413: Set document metadata failed: Setting attribute m
etadata::gedit-spell-language not supported

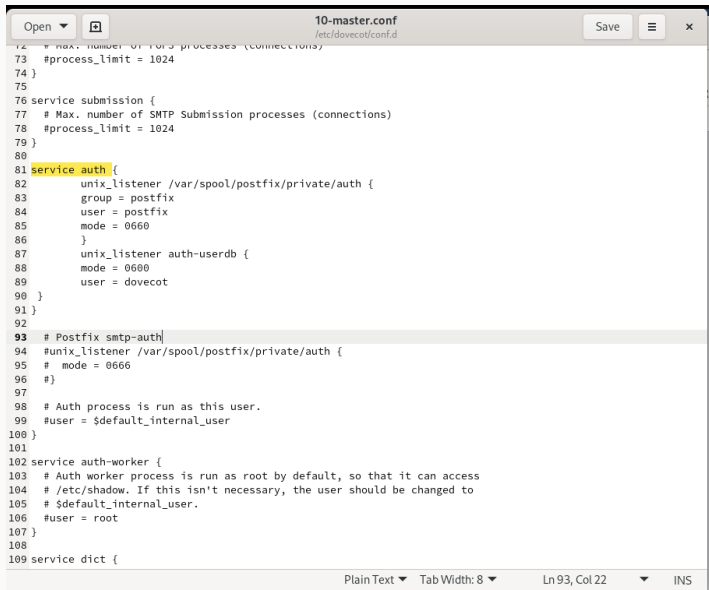
** (gedit:3411): WARNING **: 11:19:38.415: Set document metadata failed: Setting attribute m
etadata::gedit-encoding not supported

** (gedit:3411): WARNING **: 11:19:39.271: Set document metadata failed: Setting attribute m
etadata::gedit-position not supported

(gedit:3411): dconf-WARNING **: 11:19:39.291: failed to commit changes to dconf: Failed to e
xecute child process "dbus-launch" (No such file or directory)
[root@server conf.d]# systemctl restart postfix
[root@server conf.d]# systemctl restart dovecot
[root@server conf.d]# echo .| mail -s "LMTP test" user@user.net
[root@server conf.d]#
```

Рис. 4: Логи доставки через LMTP

Проверка почтового ящика Maildir



```
12 # Max. number of POP3 processes (connections)
13 #process_limit = 1024
14 }
15
16 service submission {
17 # Max. number of SMTP Submission processes (connections)
18 #process_limit = 1024
19 }
20
21 service auth {
22     unix_listener /var/spool/postfix/private/auth {
23         group = postfix
24         user = postfix
25         mode = 0660
26     }
27     unix_listener auth-userdb {
28         mode = 0600
29         user = dovecot
30     }
31 }
32
33 # Postfix smtp-auth
34 #unix_listener /var/spool/postfix/private/auth {
35 #    mode = 0666
36 #}
37
38 # Auth process is run as this user.
39 #user = $default_internal_user
40 }
41
42 service auth-worker {
43 # Auth worker process is run as root by default, so that it can access
44 # /etc/shadow. If this isn't necessary, the user should be changed to
45 # $default_internal_user.
46 #user = root
47 }
48
49 service dict {
```

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Служба auth в Dovecot

```
root@server:/etc/dovecot/conf.d

root@server:/etc/dovecot/conf.d x root@server:~ x

** (gedit:3650): WARNING **: 11:30:26.889: Set document metadata failed: Setting attribute etadata::gedit-encoding not supported

** (gedit:3650): WARNING **: 11:30:29.288: Set document metadata failed: Setting attribute etadata::gedit-spell-language not supported

** (gedit:3650): WARNING **: 11:30:29.290: Set document metadata failed: Setting attribute etadata::gedit-encoding not supported

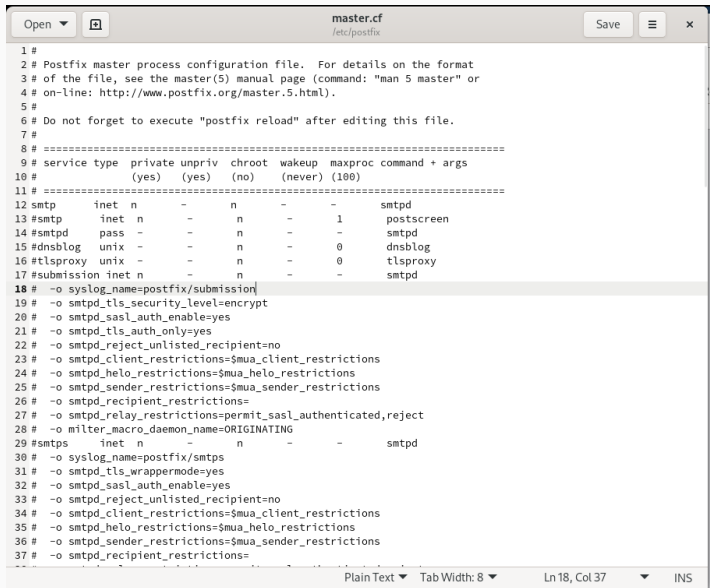
** (gedit:3650): WARNING **: 11:30:30.057: Set document metadata failed: Setting attribute etadata::gedit-spell-language not supported

** (gedit:3650): WARNING **: 11:30:30.061: Set document metadata failed: Setting attribute etadata::gedit-encoding not supported

** (gedit:3650): WARNING **: 11:30:30.867: Set document metadata failed: Setting attribute etadata::gedit-position not supported

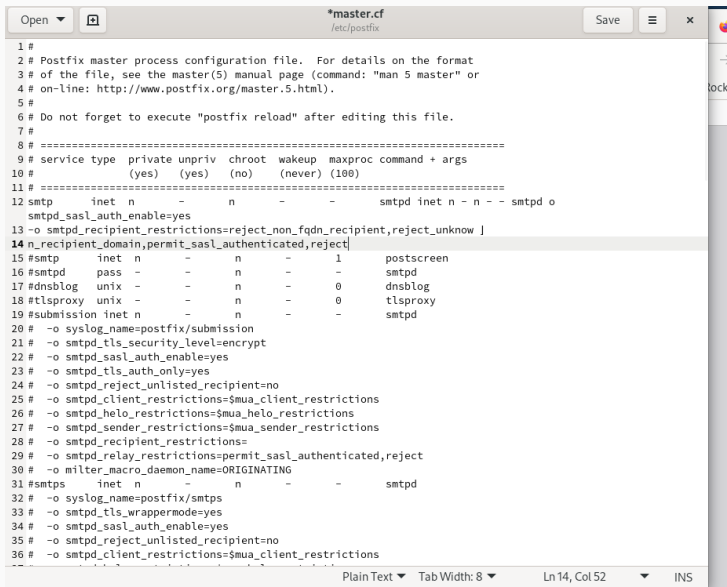
(gedit:3650): dconf-WARNING **: 11:30:30.941: failed to commit changes to dconf: Failed to execute child process "dbus-launch" (No such file or directory)
[root@server conf.d]# postconf -e 'smtpd_sasl_type = dovecot'
[root@server conf.d]# postconf -e 'smtpd_sasl_path = private/auth'
[root@server conf.d]# postconf -e 'smtpd_recipient_restrictions =
reject_unknown_recipient_domain, permit_mynetworks,
reject_non_fqdn_recipient, reject_unauth_destination,
reject_unverified_recipient, permit'
postconf: fatal: -e, -X, or -# accepts no multi-line input
[root@server conf.d]# postconf -e 'smtpd_recipient_restrictions = reject_unknown_recipient
omain, permit_mynetworks, reject_non_fqdn_recipient, reject_unauth_destination, reject_un
ified_recipient, permit'
[root@server conf.d]# postconf -e 'mynetworks = 127.0.0.0/8'
[root@server conf.d]#
```

Настройка SASL в Postfix



```
1 #
2 # Postfix master process configuration file. For details on the format
3 # of the file, see the master(5) manual page (command: "man 5 master" or
4 # on-line: http://www.postfix.org/master.5.html).
5 #
6 # Do not forget to execute "postfix reload" after editing this file.
7 #
8 # =====
9 # service type private unpriv chroot wakeup maxproc command + args
10 #          (yes)   (yes)   (no)   (never) (100)
11 # =====
12 smtp      inet n       -       n       -       -       smtpd
13 #smtp     inet n       -       n       -       1       postscreen
14 #smtpd    pass -       -       n       -       -       smtpd
15 #dnsblog  unix -       -       n       -       0       dnsblog
16 #tlsproxy unix -       -       n       -       0       tlsproxy
17 #submission inet n      -       n       -       -       smtpd
18 # -o syslog_name=postfix/submission
19 # -o smtpd_tls_security_level=encrypt
20 # -o smtpd_sasl_auth_enable=yes
21 # -o smtpd_tls_auth_only=yes
22 # -o smtpd_reject_unlisted_recipient=no
23 # -o smtpd_client_restrictions=$mua_client_restrictions
24 # -o smtpd_helo_restrictions=$mua_helo_restrictions
25 # -o smtpd_sender_restrictions=$mua_sender_restrictions
26 # -o smtpd_recipient_restrictions=
27 # -o smtpd_relay_restrictions=permit_sasl_authenticated,reject
28 # -o milter_macro_daemon_name=ORIGINATING
29 #smtps    inet n       -       n       -       -       smtpd
30 # -o syslog_name=postfix/smtps
31 # -o smtpd_tls_wrappermode=yes
32 # -o smtpd_sasl_auth_enable=yes
33 # -o smtpd_reject_unlisted_recipient=no
34 # -o smtpd_client_restrictions=$mua_client_restrictions
35 # -o smtpd_helo_restrictions=$mua_helo_restrictions
36 # -o smtpd_sender_restrictions=$mua_sender_restrictions
37 # -o smtpd_recipient_restrictions=
```

Временная конфигурация для теста SMTP AUTH



```
1 #
2 # Postfix master process configuration file. For details on the format
3 # of the file, see the master(5) manual page (command: "man 5 master" or
4 # on-line: http://www.postfix.org/master.5.html).
5 #
6 # Do not forget to execute "postfix reload" after editing this file.
7 #
8 # =====
9 # service type  private unpriv  chroot  wakeup  maxproc command + args
10 #              (yes)   (yes)   (no)   (never) (100)
11 # =====
12 smtp      inet  n       -       n       -       -       smtpd inet n - n - - smtpd o
13          smtpd_sasl_auth_enable=yes
14 -o smtpd_recipient_restrictions=reject_non_fqdn_recipient,reject_unknown ]
15 n_recipient_domain,permit_sasl_authenticated,reject|
16 #smtp      inet  n       -       n       -       1       postscreen
17 #smtpd     pass  -       -       n       -       -       smtpd
18 #dnsblog   unix  -       -       n       -       0       dnsblog
19 #tlsproxy  unix  -       -       n       -       0       tlsproxy
20 #submission inet  n       -       n       -       -       smtpd
21 # -o syslog_name=postfix/submission
22 # -o smtpd_tls_security_level=encrypt
23 # -o smtpd_sasl_auth_enable=yes
24 # -o smtpd_tls_auth_only=yes
25 # -o smtpd_reject_unlisted_recipient=no
26 # -o smtpd_client_restrictions=$mua_client_restrictions
27 # -o smtpd_helo_restrictions=$mua_helo_restrictions
28 # -o smtpd_sender_restrictions=$mua_sender_restrictions
29 # -o smtpd_recipient_restrictions=
30 # -o smtpd_relay_restrictions=permit_sasl_authenticated,reject
31 # -o milter_macro_daemon_name=ORIGINATING
32 #smtps     inet  n       -       n       -       -       smtpd
33 # -o syslog_name=postfix/smtps
34 # -o smtpd_tls_wrappermode=yes
35 # -o smtpd_sasl_auth_enable=yes
36 # -o smtpd_reject_unlisted_recipient=no
37 # -o smtpd_client_restrictions=$mua_client_restrictions
```

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Проверка SMTP AUTH

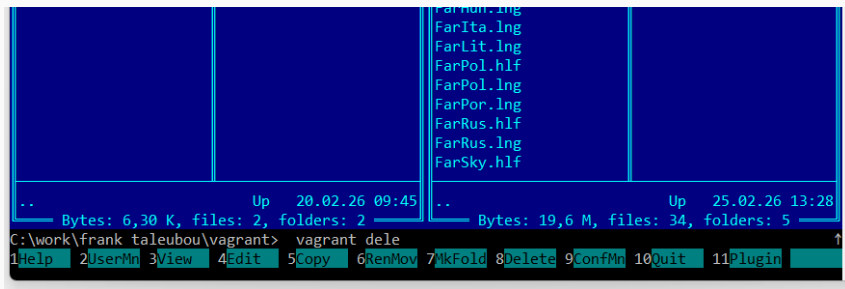


Рис. 9: Успешная SMTP-аутентификация

Подключение TLS (сертификат Dovecot)

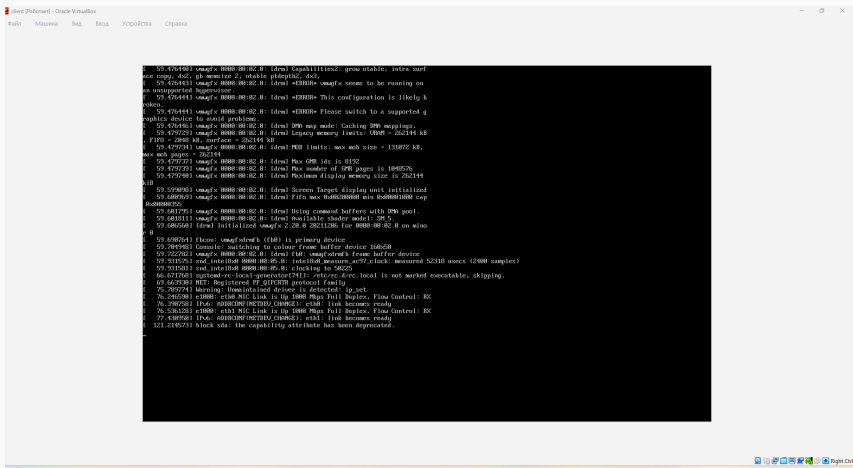


Рис. 10: TLS-параметры Postfix

Порт submission (587) и обязательный TLS

```
client [Работает] - Oracle VirtualBox
Файл  Машина  Вид  Ввод  Устройства  Справка

Activities  Terminal  Mar 11 12:10  en  [Icons]

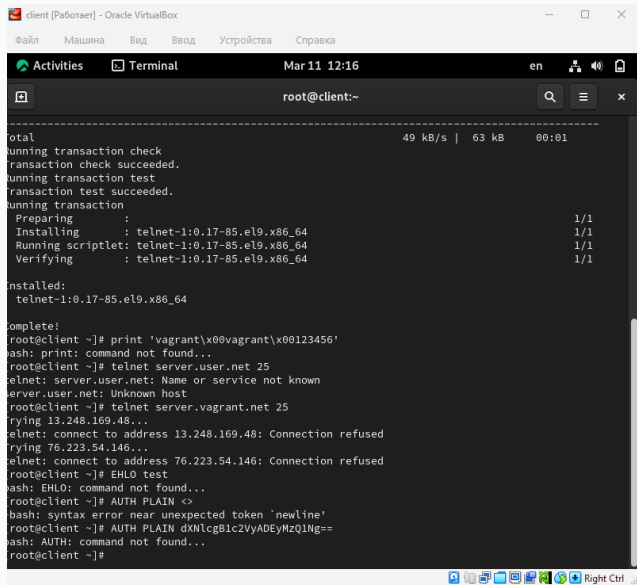
root@client:~

[vagrant@client ~]$ sudo -i
[root@client ~]# dnf -y install telnet
Extra Packages for Enterprise Linux 9 - x86_64                27 kB/s | 28 kB    00:01
Extra Packages for Enterprise Linux 9 - x86_64                204 kB/s | 20 MB   01:42
Extra Packages for Enterprise Linux 9 openh264 (From Cisco) - 1.6 kB/s | 993 B    00:00
Rocky Linux 9 - BaseOS                                         5.8 kB/s | 4.3 kB    00:00
Rocky Linux 9 - BaseOS                                         4.9 MB/s | 16 MB   00:03
Rocky Linux 9 - AppStream                                       11 kB/s | 4.8 kB    00:00
Rocky Linux 9 - AppStream                                       4.7 MB/s | 16 MB   00:03
Rocky Linux 9 - Extras                                         561 B/s | 3.1 kB    00:05
Rocky Linux 9 - Extras                                         21 kB/s | 17 kB    00:00
Dependencies resolved.
=====
Package              Architecture      Version           Repository         Size
=====
Installing:
telnet                x86_64            1:0.17-85.el9     appstream           63 k
Transaction Summary
=====
Install 1 Package

Total download size: 63 k
Installed size: 121 k
Downloading Packages:
telnet-0.17-85.el9.x86_64.rpm                                95 kB/s | 63 kB    00:00
-----
Total                                                         49 kB/s | 63 kB    00:01
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing...

```


Проверка STARTTLS и аутентификации на 587



```
client [Работаer] - Oracle VirtualBox
Файл  Машина  Вид  Ввод  Устройства  Справка

Activities  Terminal  Mar 11 12:16  en  [Icons]

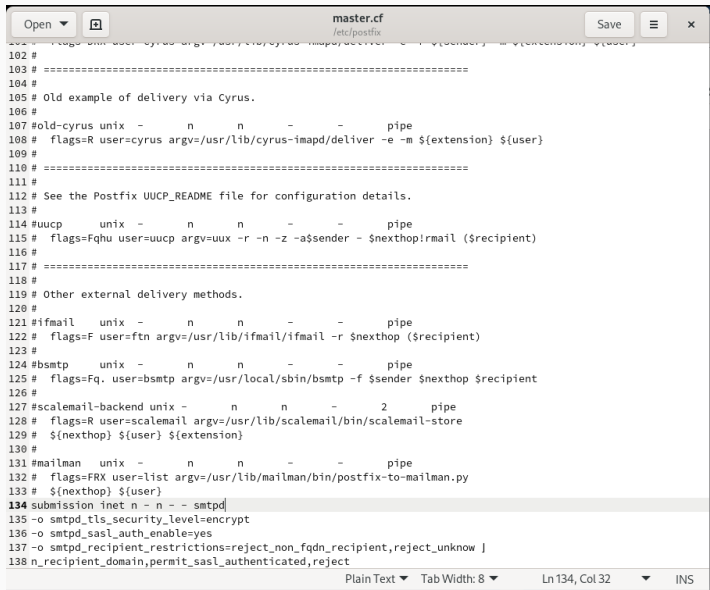
root@client:~

-----
total                               49 kB/s | 63 kB    00:01
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      :                                1/1
  Installing     : telnet-1:0.17-85.el9.x86_64    1/1
  Running scriptlet: telnet-1:0.17-85.el9.x86_64    1/1
  Verifying      : telnet-1:0.17-85.el9.x86_64    1/1

Installed:
  telnet-1:0.17-85.el9.x86_64

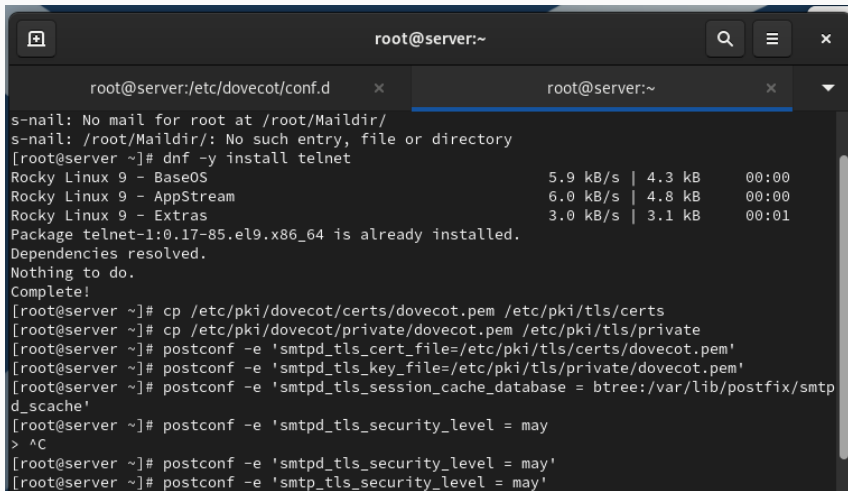
Complete!
root@client ~]# print 'vagrant\x00vagrant\x00123456'
bash: print: command not found...
root@client ~]# telnet server.user.net 25
telnet: server.user.net: Name or service not known
server.user.net: Unknown host
root@client ~]# telnet server.vagrant.net 25
Trying 13.248.169.48...
telnet: connect to address 13.248.169.48: Connection refused
Trying 76.223.54.146...
telnet: connect to address 76.223.54.146: Connection refused
root@client ~]# EHLO test
bash: EHLO: command not found...
root@client ~]# AUTH PLAIN <>
bash: syntax error near unexpected token `newline'
root@client ~]# AUTH PLAIN dXNlcm91dC5yYXZlbnQ1MzQ1Ng==
bash: AUTH: command not found...
root@client ~]#
```

Настройка Evolution



```
102 #
103 # =====
104 #
105 # Old example of delivery via Cyrus.
106 #
107 #old-cyrus unix -      n      n      -      -      pipe
108 # flags=R user=cyrus argv=/usr/lib/cyrus-imapd/deliver -e -m ${extension} ${user}
109 #
110 # =====
111 #
112 # See the Postfix UUCP_README file for configuration details.
113 #
114 #uucp      unix -      n      n      -      -      pipe
115 # flags=Fqhu user=uucp argv=uux -r -n -z -a$sender - $nexthop!rmail ($recipient)
116 #
117 # =====
118 #
119 # Other external delivery methods.
120 #
121 #ifmail    unix -      n      n      -      -      pipe
122 # flags=F user=ftn argv=/usr/lib/ifmail/ifmail -r $nexthop ($recipient)
123 #
124 #bsmtp     unix -      n      n      -      -      pipe
125 # flags=Fq. user=bsmtp argv=/usr/local/sbin/bsmtp -f $sender $nexthop $recipient
126 #
127 #scalemail-backend unix -      n      n      -      2      pipe
128 # flags=R user=scalemail argv=/usr/lib/scalemail/bin/scalemail-store
129 # ${nexthop} ${user} ${extension}
130 #
131 #mailman   unix -      n      n      -      -      pipe
132 # flags=FRX user=list argv=/usr/lib/mailman/bin/postfix-to-mailman.py
133 # ${nexthop} ${user}
134 submission inet n - n - - smtpd
135 -o smtpd_tls_security_level=encrypt
136 -o smtpd_sasl_auth_enable=yes
137 -o smtpd_recipient_restrictions=reject_non_fqdn_recipient,reject_unknown
138 n_recipient_domain,permit_sasl_authenticated,reject
```

Результат проверки



```
root@server:~  
root@server:/etc/dovecot/conf.d x root@server:~ x  
s-nail: No mail for root at /root/Maildir/  
s-nail: /root/Maildir/: No such entry, file or directory  
[root@server ~]# dnf -y install telnet  
Rocky Linux 9 - BaseOS 5.9 kB/s | 4.3 kB 00:00  
Rocky Linux 9 - AppStream 6.0 kB/s | 4.8 kB 00:00  
Rocky Linux 9 - Extras 3.0 kB/s | 3.1 kB 00:01  
Package telnet-1:0.17-85.el9.x86_64 is already installed.  
Dependencies resolved.  
Nothing to do.  
Complete!  
[root@server ~]# cp /etc/pki/dovecot/certs/dovecot.pem /etc/pki/tls/certs  
[root@server ~]# cp /etc/pki/dovecot/private/dovecot.pem /etc/pki/tls/private  
[root@server ~]# postconf -e 'smtpd_tls_cert_file=/etc/pki/tls/certs/dovecot.pem'  
[root@server ~]# postconf -e 'smtpd_tls_key_file=/etc/pki/tls/private/dovecot.pem'  
[root@server ~]# postconf -e 'smtpd_tls_session_cache_database = btree:/var/lib/postfix/smtp  
d_scache'  
[root@server ~]# postconf -e 'smtpd_tls_security_level = may  
> ^C  
[root@server ~]# postconf -e 'smtpd_tls_security_level = may'  
[root@server ~]# postconf -e 'smtp_tls_security_level = may'
```

Рис. 14: Сообщения в Evolution

Логи сервера при работе TLS и AUTH

```
root@server:~
root@server:/etc/dovecot/conf.d
root@server:~

ix mdns memcache minidlna mongodb mosh mountd mqtt mqtt-tls ms-wbt mssql murmur mysql nbd ne
bula netbios-ns netdata-dashboard nfs nfs3 nmea-0183 nrpe ntp nut openvpn ovirt-imageio ovir
t-storageconsole ovirt-vmconsole plex pmcd pmproxy pmwebapi pmwebapis pop3 pop3s postgresql
privoxy prometheus prometheus-node-exporter proxy-dhcp ps2link ps3netsrv ptp pulseaudio pupp
etmaster quassel radius rdp redis redis-sentinel rpc-bind rquotad rsh rsyncd rtsp salt-maste
r samba samba-client samba-dc sane sip sips slp smtp smtp-submission smtps snmp snmptls snmp
tls-trap snmptrap spideroak-lansync spotify-sync squid ssdp ssh ssh-custom steam-streaming s
vdrp svn syncthing syncthing-gui syncthing-relay synergy syslog syslog-tls telnet tentacle t
ftp tile38 tinc tor-socks transmission-client upnp-client vdsn vnc-server warpinator wbm-ht
tp wbm-https wireguard ws-discovery ws-discovery-client ws-discovery-tcp ws-discovery-udp w
sman wsmans xdmcp xmpp-bosh xmpp-client xmpp-local xmpp-server zabbix-agent zabbix-server ze
rotier
[root@server ~]# firewall-cmd --add-service=smtp-submission
success
[root@server ~]# firewall-cmd --add-service=smtp-submission --permanent
success
[root@server ~]# firewall-cmd --reload
success
[root@server ~]# systemctl restart postfix
Job for postfix.service failed because the control process exited with error code.
See "systemctl status postfix.service" and "journalctl -xeu postfix.service" for details.
[root@server ~]# openssl s_client -starttls smtp -crlf -connect server.vagrant.net:587
801B44331D7F0000:error:8000006F:system library:BI0_connect:Connection refused:crypto/bio/bio
_sock2.c:114:calling connect()
801B44331D7F0000:error:10000067:BI0 routines:BI0_connect:connect error:crypto/bio/bio_sock2.
c:116:
801B44331D7F0000:error:8000006F:system library:BI0_connect:Connection refused:crypto/bio/bio
_sock2.c:114:calling connect()
801B44331D7F0000:error:10000067:BI0 routines:BI0_connect:connect error:crypto/bio/bio_sock2.
c:116:
connect:errno=111
[root@server ~]#
```

Итоги работы

- Настроена доставка почты через LMTP (Postfix → Dovecot)
- Реализована SMTP-аутентификация пользователей (SASL через Dovecot)
- Включён SMTP over TLS и настроен submission 587 с обязательным шифрованием
- Выполнено тестирование через консоль и Evolution
- Конфигурация перенесена в provisioning, обеспечено воспроизводимое развёртывание